

CPSC 304 Project Cover Page

Milestone #: ____1____

Date: ____Feb 1st, 2026____

Group Number: ____90____

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Description:

1. Domain of the application:

The domain of our application is law firm case management. It represents the day-to-day operational aspects of a legal practice, including the cases a lawyer is handling, the work performed over time, and the progression of cases through hearings and courthouses. It also encompasses the tracking of evidence associated with each case and hearing, as well as the billing process upon case completion.

2. What aspects of the domain are modeled by the database.

- **Client and Case Structure**

We store information about clients and the cases they bring to the firm. Each case is associated with exactly one client, as every case must belong to a lawyer through that client relationship. A client may exist in the system even if they do not currently have any active cases.

- **Lawyer Assignment and Practice Areas**

Each case is assigned one primary lawyer to ensure clear accountability. Lawyers can exist in the system even if they are not currently assigned to any cases. We also model that each lawyer practices in exactly one area—Family, Criminal, or Corporate—so the system can support different billing rates and rules based on the type of law practiced.

- **Work Logs and Case Effort Tracking**

Lawyers record their work through work logs. Each work log is linked to exactly one lawyer and one case, allowing the system to track effort over time and enabling accurate billing once the case is complete.

- **Billing and Invoicing**

Billing occurs only after a case has been completed. As a result, an open case may not have an invoice, but once it is closed, it can generate a final invoice. This reflects common law firm billing practices while keeping the data model straightforward and realistic.

- **Hearings and Courthouses**

Hearings are dependent on cases, as a hearing cannot exist without an associated case. For this reason, hearings are treated as weak entities, identified by the case and the hearing date. Each hearing must take place at exactly one courthouse, since a hearing requires a physical location.

- **Evidence Management**

Evidence is associated with a case rather than directly with a hearing, because evidence is gathered for the overall case. Evidence may later be presented at one or more hearings, or not presented at all. This is why evidence and hearings are modeled as a many-to-many relationship.

The system models core law firm operations by capturing clients, cases, lawyers, and their relationships. Each case belongs to a single client and is managed by one primary lawyer, who

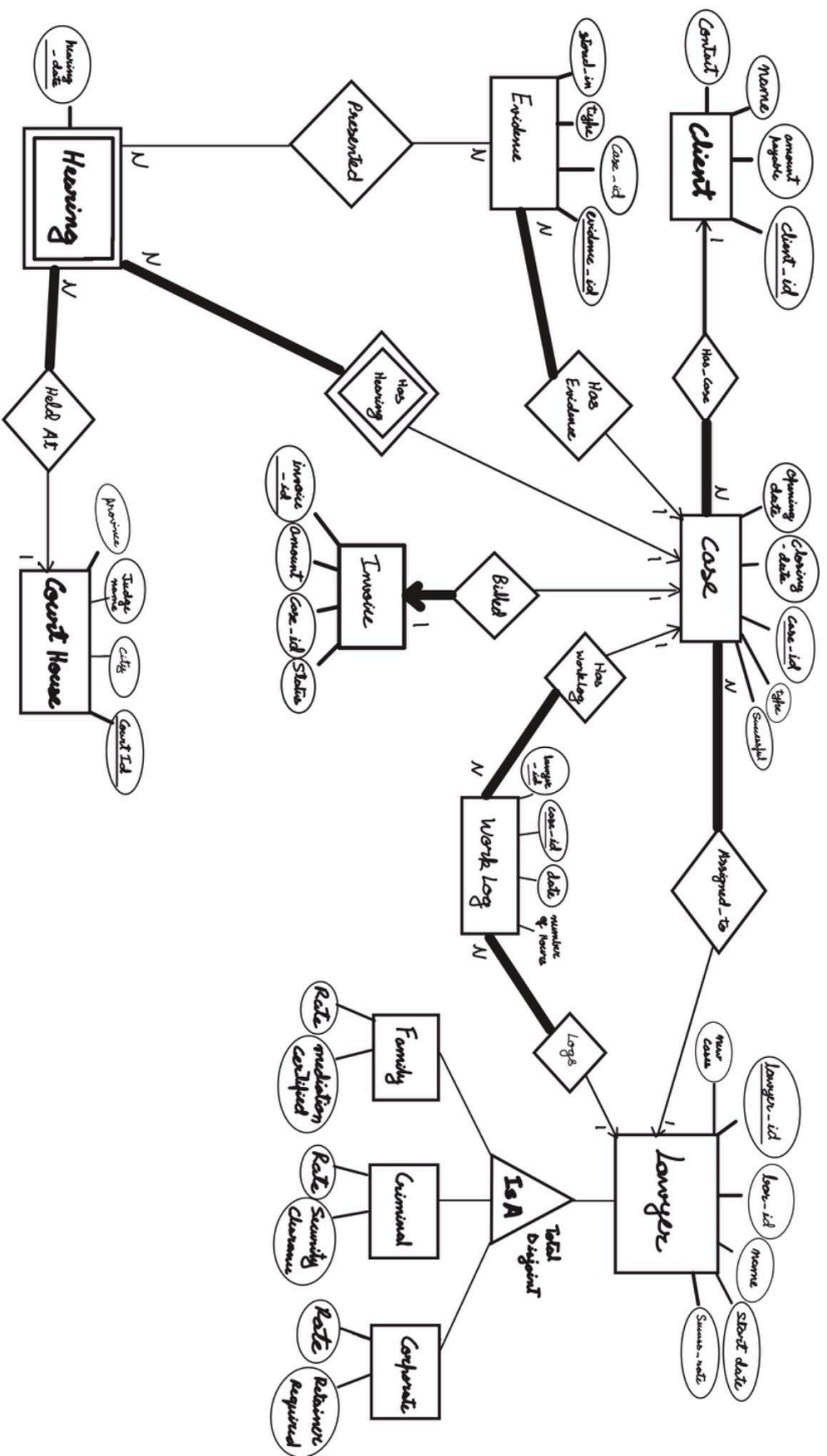
practices in a specific area of law. Lawyers track their work through work logs linked to cases, enabling accurate billing once a case is closed. The model also represents hearings as case-dependent events held at courthouses and manages evidence at the case level, allowing it to be associated with multiple hearings or none at all.

3. ER Diagram

See picture attached on the next page.

4. Declaration of AI usage

We acknowledge that AI (ChatGPT) was used in rewording this document. A pdf of the conversation is attached at the end of this file.



ER - Diagram

The domain of our application is law firm case management. It models the day to day operational side of a law firm, what cases the lawyer is working on, what work gets done over time, and how cases move through hearings and courthouses. It also covers how the lawyer tracks evidence tied to a case & hearing, and how billing happens when the case is finished.

i need you to simply reword this in a more articulate manner. dont chnag ethe content of the paragraph. omly the wording to sound more professional and articulate

The domain of our application is law firm case management. It represents the day-to-day operational aspects of a legal practice, including the cases a lawyer is handling, the work performed over time, and the progression of cases through hearings and courthouses. It also encompasses the tracking of evidence associated with each case and hearing, as well as the billing process upon case completion.



We store clients and the cases they bring to the firm.
Each case is owned by exactly one client, because a case

has to belong to a lawyer. A client can exist even if they currently have no active cases.

Every case has one primary lawyer assigned to it, since someone has to be accountable for the case. Lawyers can exist in the system even if they are not currently assigned to any cases. We also model that each lawyer works in exactly one area, Family, Criminal, or Corporate, so the system can store different rate details depending on the type of law they practice.

Lawyers record the work they do using work logs. Each work log is tied to one specific lawyer and one specific case. This makes it easy to see how effort adds up over time and supports accurate billing later.

We only bill once the case is finished. That is why a case may not have an invoice while it is still open, but once it closes it can produce a final invoice. This matches how many firms handle billing, and keeps the model simple and realistic.

Hearings are dependent on cases. A hearing does not make sense without a case attached to it, so we treat it as a weak entity identified by the case and the hearing date. Every hearing must happen at one courthouse, since you cannot have a hearing without a location.

Evidence is tied to a case, not directly to a hearing, because evidence is collected for the whole case. Later,

evidence can be presented at hearings, or never presented at all. That is why we model evidence and hearings as many to many.

now do the same for these paragraphs as well as add small subheadings for each of them. highlight the subheadings too

Client and Case Structure

We store information about clients and the cases they bring to the firm. Each case is associated with exactly one client, as every case must belong to a lawyer through that client relationship. A client may exist in the system even if they do not currently have any active cases.

Lawyer Assignment and Practice Areas

Each case is assigned one primary lawyer to ensure clear accountability. Lawyers can exist in the system even if they are not currently assigned to any cases. We also model that each lawyer practices in exactly one area—Family, Criminal, or Corporate—so the system can support different billing rates and rules based on the type of law practiced.

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Hearings and Courthouses

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Evidence Management

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could you also provide a small summary of the text I pasted in the second prompt only.

Summary

The system models core law firm operations by capturing clients, cases, lawyers, and

their relationships. Each case belongs to a single client and is managed by one primary lawyer, who practices in a specific area of law. Lawyers track their work through work logs linked to cases, enabling accurate billing once a case is closed. The model also represents hearings as case-dependent events held at courthouses and manages evidence at the case level, allowing it to be associated with multiple hearings or none at all.

